

# Chutong Yang

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## Research Focus

My research focuses on learning theory and algorithms for reliable machine learning, with an emphasis on calibration and optimization. My recent work explores LLM reasoning and multi-agent collaboration, including the design and evaluation of AI agents for automated research.

## Education

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|---------------------------------------------------------------------------------------|---------------------|
| <b>The University of Texas at Austin</b> , Ph.D. in Computer Science                  | Aug 2023 – Present  |
| • Recipient, <b>Amazon AI PhD Fellowship</b>                                          |                     |
| <b>Stanford University</b> , M.S. in Computer Science                                 | Sep 2021 – Jun 2023 |
| • Full financial support for two years.                                               |                     |
| <b>University of California, San Diego</b> , B.S. in Computer Science and Mathematics | Sep 2016 – Jun 2020 |

## Experience

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Amazon Web Services (AWS)</b> , Applied Scientist Intern                                                                                                                            | Summer 2026        |
| • Built TCSAlgBench, a benchmark of sandboxed, research-level theoretical computer science challenges for evaluating LLMs and agentic systems.                                         |                    |
| • Designed and evaluated agent harnesses, including plan-and-execute and Monte Carlo tree search (MCTS), finding that agentic planning performed best among the evaluated approaches.  |                    |
| <b>The University of Texas at Austin</b> , Research Assistant                                                                                                                          | Aug 2023 – Present |
| • Developed optimal online recalibration algorithms, with simultaneous calibration and squared-loss improvement guarantees for discretized forecasts under adversarial data sequences. |                    |
| • Develop methods for reliable LLM reasoning through uncertainty-guided multi-agent deliberation and RL-based confidence calibration.                                                  |                    |
| <b>Stanford University</b> , Research Assistant                                                                                                                                        | 2022 – 2023        |
| • Researched algorithmic fairness with Prof. Omer Reingold, focusing on multicalibration.                                                                                              |                    |
| • Developed optimization algorithms with Prof. Aaron Sidford using interior-point methods.                                                                                             |                    |

## Skills

**Programming:** Python, C++, Java, MATLAB   **ML:** PyTorch

## Publications

(Authors are listed alphabetically unless otherwise indicated by \*.)

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>13. TCSAlgBench: Benchmarking Automated Proving for Research-Level Theoretical Computer Science</b>                                                                                                                                     | Preprint, 2026           |
| <b>Chutong Yang*</b> , Xiyuan Zhang, Yu Huang, Boran Han, Soonho Kong, Shuai Zhang, Vihang Prakash Patil, Zhen Han, Michael Bohlke-Schneider, Bernie Wang<br><a href="http://arxiv.org/abs/2609.35606">http://arxiv.org/abs/2609.35606</a> |                          |
| <b>12. Gap-Free Streaming PCA Beyond Rank-One Updates: Near-Optimal Rates and Applications to Differential Privacy</b>                                                                                                                     | Preprint, 2026           |
| Anming Gu, Syamantak Kumar, Kevin Tian, <b>Chutong Yang</b><br><a href="https://arxiv.org/abs/2609.26508">https://arxiv.org/abs/2609.26508</a>                                                                                             |                          |
| <b>11. Optimal Recalibration of an Online Predictor</b>                                                                                                                                                                                    | NeurIPS 2026 (spotlight) |
| Lunjia Hu, Kevin Tian, <b>Chutong Yang</b><br><a href="https://arxiv.org/abs/2607.19689">https://arxiv.org/abs/2607.19689</a>                                                                                                              |                          |

- 10. Simultaneous Blackwell Approachability and Applications to Multiclass Omniprediction** COLT 2026  
Lunjia Hu, Kevin Tian, *Chutong Yang*  
<https://arxiv.org/abs/2602.17577>
- 9. Spherical Leech Quantization for Visual Tokenization and Generation** CVPR 2026 (Highlight)  
Yue Zhao\*, Hanwen Jiang, Zhenlin Xu, *Chutong Yang*, Ehsan Adeli, Philipp Krähenbühl  
<https://arxiv.org/abs/2512.14697>
- 8. Proportionality from Low-Dimensional Approval Data** AAMAS 2026  
Zhiyi Huang, Gregory Kehne, *Chutong Yang*  
Extended abstract: <https://doi.org/10.65109/YHKI7695>
- 7. Private Geometric Median in Nearly-Linear Time** NeurIPS 2025  
Syamantak Kumar, Daogao Liu, Kevin Tian, *Chutong Yang*  
TPDP 2025 Workshop  
<https://arxiv.org/abs/2505.20189>
- 6. Omnipredicting Single-Index Models with Multi-Index Models** STOC 2025  
Lunjia Hu, Kevin Tian, *Chutong Yang*  
<https://arxiv.org/abs/2411.13083>
- 5. Testing Calibration in Nearly-Linear Time** NeurIPS 2024  
Lunjia Hu, Arun Jambulapati, Kevin Tian, *Chutong Yang*  
<https://arxiv.org/abs/2402.13187>
- 4. Omnipredictors for Constrained Optimization** ICML 2023  
Lunjia Hu, Inbal Livni-Navon, Omer Reingold, *Chutong Yang*  
<https://arxiv.org/abs/2209.07463>
- 3. Active Learning Polynomial Threshold Functions** NeurIPS 2022  
Omri Ben-Eliezer, Max Hopkins, *Chutong Yang*, Hantao Yu  
<https://arxiv.org/abs/2201.09433>
- 2. A Fast Exact Algorithm for Deployment of Sensor Nodes for Internet of Things** ISF 2020  
Qinghua Zheng\*, *Chutong Yang*, Haijun Yang, Jianhe Zhou  
Information Systems Frontiers: <https://link.springer.com/article/10.1007/s10796-018-9890-3>
- 1. Detailed placement for IR drop mitigation by power staple insertion in sub-10nm VLSI** DATE 2019  
Sun ik Heo, Andrew B Kahng, Minsoo Kim, Lutong Wang, *Chutong Yang*  
<https://ieeexplore.ieee.org/abstract/document/8715096>

## Professional Service

Conference reviewer: NeurIPS 2024, ICLR 2025, ICML 2025, ITCS 2026, NeurIPS 2026